7th International Workshop on Active Inference | Madrid, 14–16 October 2026

Return Is Not Enough

Instruments for dissecting what each term of the expected free energy actually does

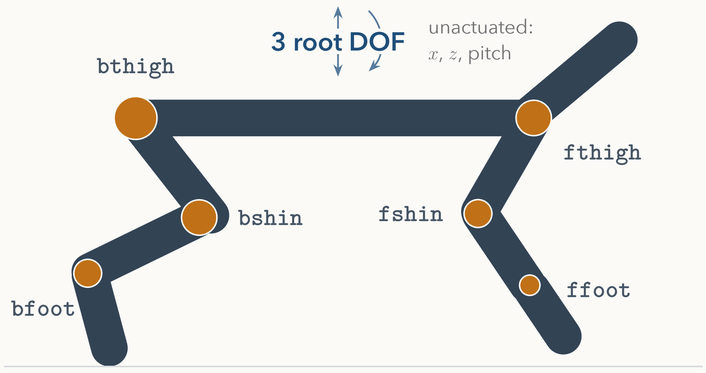
Stefano Goria | AIM Research Lab | stefano@aimresearchlab.com

github.com/egolabs-ai/iwai26

paper · data · code

1 THE TESTBED

A planar cheetah, two rewards with different geometry, and two ways to push the agent off the distribution it trained on.



Six hinges about the pitch axis, dot area $\propto$ actuator gear. Limits run from $\pm 28°$ at `ffoot` to $-230°$ to $50°$ at `bfoot`.

WHAT THE AGENT SEES AND DOES

observation	17-D	$=$ `qpos[1:]` $\oplus$ `qvel`
action	6-D torque	$\in [-1, 1]^6$, gear-scaled

TWO REWARDS, TWO GEOMETRIES

`cheetah-run`	`cheetah-flip`
forward velocity, clipped at $v^\star = 10$	angular momentum about pitch
degenerate: many gaits reach the same R	**not degenerate**: rotation fixes the body

Both: 17-D observation, 6-D action, $b = 3 \times 10^4$ frames, 5 seeds.

PUSHING THE AGENT OFF ITS DISTRIBUTION



$\theta = 0°$
the world it trained in

$\theta = -25°$
same agent, no retraining

Rotating gravity by θ is the same as running up a slope of θ. We sweep θ from $0°$ to $-40°$, and separately inject i.i.d. action noise up to $\sigma = 0.5$. Both are applied to the **frozen final checkpoint**.

THE LINE-UP

Three agents at matched parameter count (within $\pm 7\%$) and matched budget: **CEM-AI** at $\lambda = 0.01$ and $\lambda = 0$ ($E{=}15$ ensemble, 5.33M params, CEM with $N{=}700$, $H{=}15$), and **TD-MPC2** ($\approx$ 5M).

The canonical "$10\times$ more sample-efficient than SAC" claim compares against one dashed horizontal line: the source figure plots no SAC learning curve, and stops the AI agent at 100 episodes. We use a modern baseline instead, and an **oracle CEM** (true physics, true reward, no learning) to separate planner geometry from model error. It peaks at $v_x \approx 6.8$, below $v^\star$.

2 RETURN IS DEGENERATE

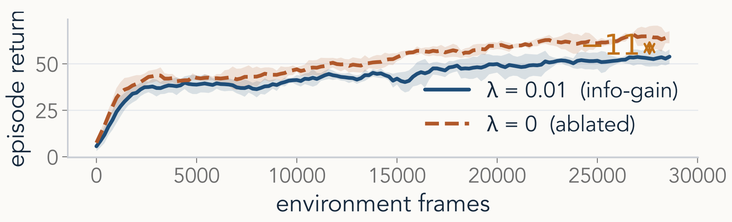
A scalar score collapses the trajectory. On a reward with many equal-scoring solutions it cannot see what the epistemic term changed.

THE TERM UNDER TEST

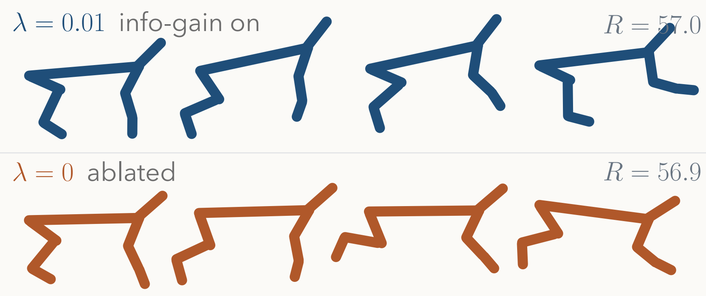
$$G_\lambda(\tau) = \underbrace{G_{\text{ext}}(\tau)}_{\text{reward}} + \lambda \underbrace{G_{\text{epi}}(\tau)}_{\text{information gain}}$$

$$\text{IG}(s,a) = \mathcal{H}\Big[\tfrac{1}{E}\textstyle\sum_e p_e(\cdot \mid s,a)\Big] - \tfrac{1}{E}\textstyle\sum_e \mathcal{H}[p_e(\cdot \mid s,a)]$$

$\lambda \in \{0,\ 0.01\}$ is the only difference between the two agents.



What return says. The bonus *costs* 11 points on flat ground, and the seed spread is wide enough to cover that difference. Read either way, return says to drop the term.



Why it cannot see. The matched-return rollouts closest to $R = 57$, drawn from the committed `qpos`. Both score the same; the gaits differ.

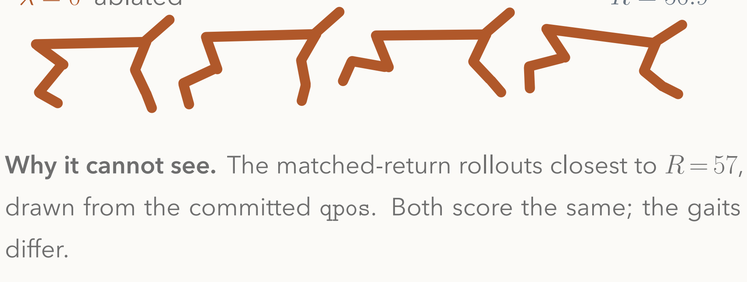
INSTRUMENT · PATH SIGNATURES

Each rollout is a path $X : [0, T] \to \mathbb{R}^{17}$. Its depth-2 signature is the 306-D vector of displacements and iterated double integrals $\int_{s<t} \mathrm{d}X_s^{(i)}\, \mathrm{d}X_t^{(j)}$, which carry the lead–lag structure of a gait.

Train on one seed per agent and test on the held-out four, rotated. A surviving axis is seed-invariant.

caught: 0.745 on `run`, 0.52 on `flip`

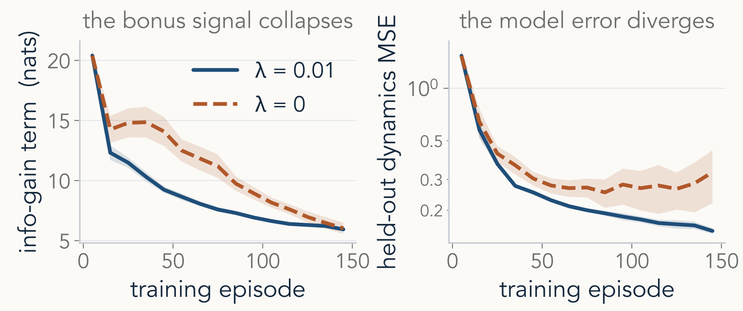
chance $= 0.5$; within-agent seeds stay distinguishable on both tasks



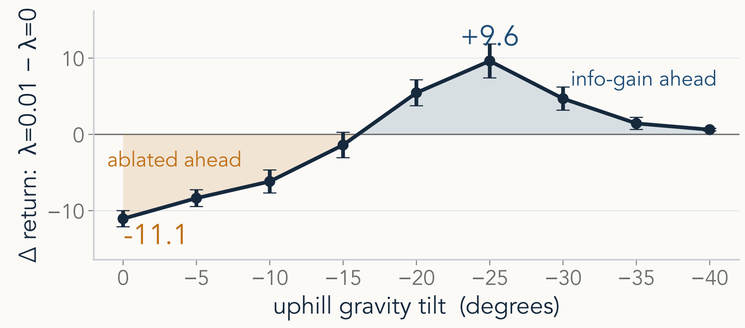
What the gait costs. More effort at equal forward velocity. $n = 131$ vs 52 rollouts, Mann–Whitney $p \le 0.005$ on all three.

3 WHAT THE BONUS BUYS

Three more instruments, and the task property that decides whether any of them fire.



Log the bonus and the model separately. The bonus term falls to the same ≈ 6 nats for both λ, while held-out dynamics error diverges anyway. The gap is widest where the signal has gone. Neither log shows this on its own.



Test where the policy does not go. Paired rollouts from the frozen checkpoints: 5 seeds $\times$ 20 shared initial conditions per tilt. The deficit reverses sign between $-15°$ and $-20°$. Past $-30°$ both agents are near zero (2.0 vs 1.4 at $-40°$), so the gap closes for want of return to differ over. Action noise keeps the agent on its manifold, and shows nothing.

THE GATING CONDITION

	`run` degenerate	`flip` not degenerate
gait axis (chance 0.5)	0.745	0.52
effort at matched R	$+10$ **to** $+34\%$	lower
held-out model error	2.7–$5\times$ **better**	$1.4\times$, unstable
best uphill ΔR	$+9.6$	-4 to -8

Every instrument that fires on `run` goes quiet, or reverses, on `flip`. That fits a bonus acting by **selection inside the equal-return set**, which has to be wide for it to have any room. We have no operational measure of width, so we report the two-task pattern and stop short of claiming the mechanism.

THE WIDER PROGRAMME

The rebundling hypothesis proposes building an agent under **one free-energy objective** and testing it by ablation: drop a component, lose a signature. This poster runs that protocol on one term of the EFE, in continuous control.

One rung is still missing. Here G_{ext} is a learned reward predictor rather than a **homeostatic prior** over preferred observations, and ablating that is the next experiment.

WHAT TO LOG IF YOU SCALE THIS

1 Evaluate at matched return, with seed-transfer discrimination and an off-manifold shift.

2 Log model accuracy, reward, the bonus term and predictive variance separately. They track different failure modes.

3 Expect to pay on-manifold for an off-manifold gain, and budget for it.

SELECTED REFERENCES

Chua et al. (2018). Deep RL in a handful of trials. *NeurIPS*. (PETS.)

Friston et al. (2017). Active inference: a process theory. *Neural Comput.*

Haarnoja et al. (2018). Soft Actor-Critic. *ICML*.

Hansen et al. (2024). TD-MPC2. *ICLR*.

Kidger et al. (2019). Deep signature transforms. *NeurIPS*.

Millidge et al. (2020). Whence the expected free energy? *Neural Comput.*

Pathak et al. (2019). Self-supervised exploration via disagreement. *ICML*.

Reizenstein & Graham (2020). The `iisignature` library. *ACM TOMS*.

Sekar et al. (2020). Planning to explore via self-supervised world models. *ICML*.

Tassa et al. (2018). DeepMind Control Suite. arXiv:1801.00690.

Tobin et al. (2017). Domain randomization. *IROS*.

Tschantz et al. (2020). Reinforcement learning through active inference. *ICLR W/S*.